

Ashish Gautam

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EDUCATION

Motilal Nehru National Institute of Technology Allahabad <i>B.Tech in Computer Science and Engineering</i>	CPI: 8.12/10 2023 – Present
B.G.B. Braj Education Academy, Mahawan, Mathura (UP) <i>Class XII – Central Board of Secondary Education (CBSE)</i>	Percentage: 87.2% 2022
B.G.B. Braj Education Academy, Mahawan, Mathura (UP) <i>Class X – Central Board of Secondary Education (CBSE)</i>	Percentage: 81.3% 2020

PROJECTS

Rubik's Cube Solver GitHub	2026
• Technologies: C++, Raylib, CMake, OpenGL, Multithreading, Graph Search Algorithms (BFS, DFS, IDA*) • Architected a highly performant C++ Rubik's Cube engine by encoding 3D state permutations into 64-bit integer bitboards, reducing complex state transitions to single-cycle bitwise operations • Engineered an Iterative Deepening A* (IDA*) AI solver guided by a custom Breadth-First Search (BFS) Corner Pattern Database, compressing graph heuristic data using 4-bit NibbleArrays to optimize RAM utilization • Built an interactive multi-threaded 3D visualizer using Raylib and OpenGL; decoupled the intensive AI solver into background threads via std::async to guarantee a stutter-free 60 FPS render loop with dynamic mouse raycasting and matrix-transformed slice animations	
Doctor Appointment System GitHub Demo	2025
• Technologies: React 19, Node.js, Express.js, MongoDB, Tailwind CSS, Vite, JWT, Redis, Razorpay, Cloudinary, Multer, Axios • Developed a MERN-based healthcare platform with dedicated Patient, Doctor, and Admin portals; implemented JWT authentication, role-based REST APIs, and secure appointment management • Integrated Gemini AI for symptom-based specialist recommendations and an AI healthcare chatbot; optimized performance using Redis caching, Cloudinary media storage, and Razorpay payment integration • Structured a scalable MVC backend with MongoDB data modeling, appointment snapshots, and responsive React SPA using Context API, React Router, and Axios	
Legacy Trunk GitHub	2024 – 2025
• Technologies: React.js, Node.js, Express.js, MongoDB, Tailwind CSS, JWT, Mongoose, AWS S3 • Engineered a recursive family tree using MongoDB \$graphLookup and Mongoose pre-save hooks for real-time traversal and dynamic multi-generational hierarchy calculation • Architected a “Time Capsule” feature with Node.js cron jobs for scheduled multimedia delivery; leveraged MongoDB TTL indexes for efficient event-driven notification management • Built a “Personal Vault” on AWS S3 with bcrypt + JWT-based RBAC; created a RESTful version-control API with cryptographic claim codes for secure user onboarding	

TECHNICAL SKILLS

Languages: C, C++, Java, JavaScript, SQL
Frameworks & Libraries: React.js, Node.js, Express.js, TailwindCSS, Mongoose
Databases: MongoDB, MySQL
Developer Tools: Git, GitHub, Visual Studio Code
Coursework: Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, DBMS, Computer Networks, Software Engineering
Areas of Interest: Full-Stack Web Development, Competitive Programming, Database Management

POSITIONS OF RESPONSIBILITY

Event Coordinator, Footprint Committee (Fashion & Ramp Walk) – Culrav, MNNIT Allahabad 2025

ACHIEVEMENTS

LeetCode (iam_ashish): Rating **1950** | Knight | Solved **600+** problems
Codeforces (ashishsps2034): Rating **1255** | Pupil | Best Rank **2187** in Round #1114 (Div. 3)
GeeksforGeeks (ashishd9ur): Solved **160+** problems across core DSA topics
CodeSparks: Secured **Top 5** position in the institute-level coding contest 2024